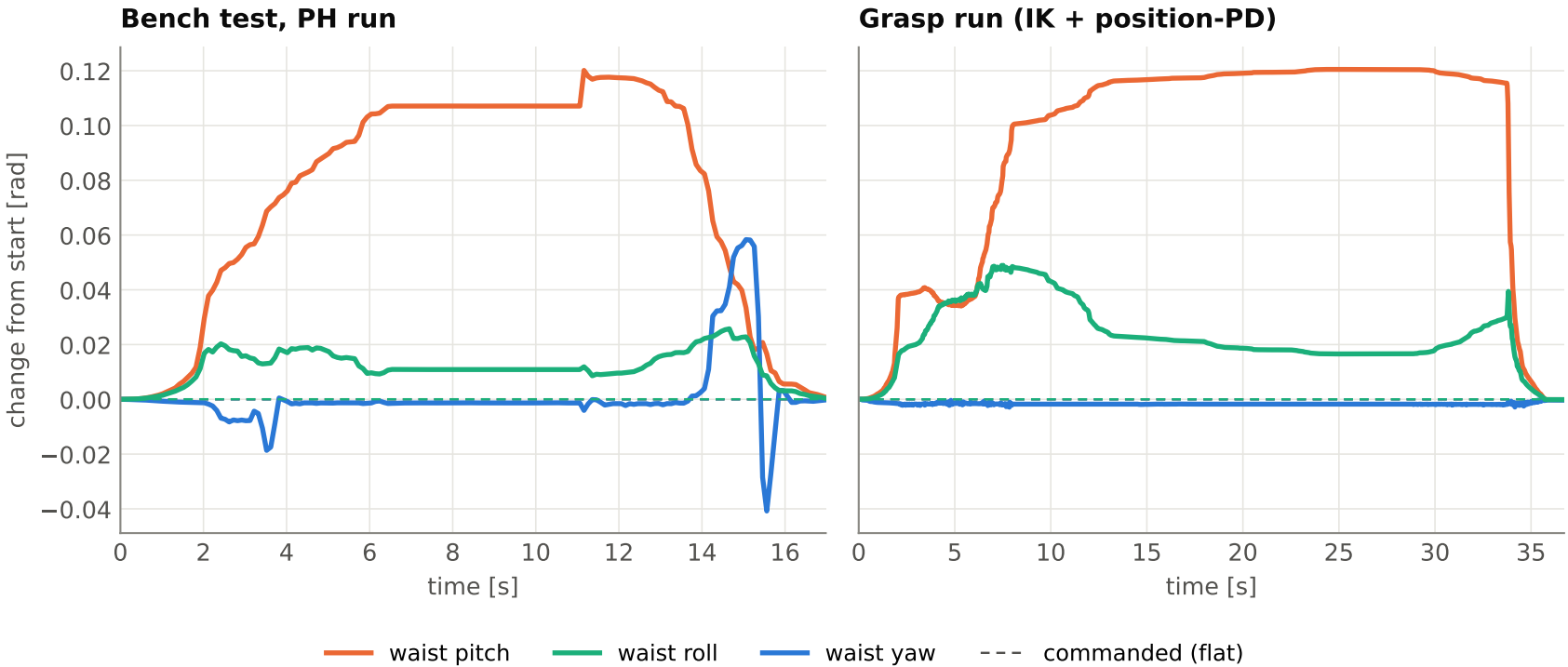
The waist is not static

Waist pitch leans up to 0.12 rad forward while the arm reaches out, although its command never changes.
Proposition 1 assumes these joints are fixed, and g(q) depends on them.